

Hypothesis Testing

Question 1: What is a null hypothesis (H_0) and why is it important in hypothesis testing?

A **Null Hypothesis (H_0)** is a statistical statement that assumes **there is no effect, no difference, or no relationship** between variables. It represents the default assumption that any observed difference in the sample data occurred **by random chance**.

In hypothesis testing, researchers start by assuming that the null hypothesis is true. The sample data is then analyzed to determine whether there is enough statistical evidence to **reject the null hypothesis**.

For example, if a company claims that the average time to resolve customer complaints is **10 minutes**, the null hypothesis would be:

$H_0: \mu = 10 \text{ minutes}$

This means the average complaint resolution time is equal to 10 minutes.

The null hypothesis is important because it provides a **baseline or reference point** for statistical testing. It allows researchers to apply statistical methods to determine whether the observed results are statistically significant or simply due to random variation.

If the evidence from the data is strong enough, the null hypothesis is rejected in favor of the **alternative hypothesis (H_1)**. Otherwise, the null hypothesis is not rejected.

Thus, the null hypothesis plays a crucial role in making **objective and data-driven decisions** in research and statistical analysis.

Question 2: What does the significance level (α) represent in hypothesis testing?

The **significance level (α)** is the probability of rejecting the null hypothesis when it is actually true. In other words, it represents the **maximum acceptable risk of making a Type I error**.

The significance level is chosen before conducting the hypothesis test and serves as a **threshold for decision making**. If the calculated p-value is less than the significance level, the null hypothesis is rejected.

Common values of significance level are:

- **0.05 (5%)**
- **0.01 (1%)**
- **0.10 (10%)**

For example, if the significance level is **0.05**, it means that there is a **5% chance of rejecting a true null hypothesis**.

The significance level helps researchers determine whether the results obtained from a sample are statistically significant. A lower significance level means that stronger evidence is required to reject the null hypothesis.

Thus, the significance level plays an important role in controlling the **probability of making incorrect conclusions** in hypothesis testing.

Question 3: Differentiate between Type I and Type II errors.

In hypothesis testing, errors may occur when making decisions about the null hypothesis. These errors are known as **Type I error** and **Type II error**.

A **Type I error** occurs when the null hypothesis is rejected even though it is actually true. This means that the test incorrectly indicates that a significant effect or difference exists when it does not.

The probability of committing a Type I error is represented by the **significance level (α)**.

Example:

A medical test concludes that a person has a disease even though the person is actually healthy.

A **Type II error** occurs when the null hypothesis is not rejected even though it is false. This means that the test fails to detect a real effect or difference.

The probability of a Type II error is represented by β .

Example:

A medical test indicates that a person is healthy even though the person actually has the disease.

Key Difference:

Type	Meaning
Type I Error	Rejecting a true null hypothesis
Type II Error	Not rejecting a false null hypothesis

Both types of errors are important considerations in statistical decision making.

Question 4: Explain the difference between a one-tailed and two-tailed test. Give an example of each.

In hypothesis testing, tests can be classified as **one-tailed tests** or **two-tailed tests**, depending on the direction of the alternative hypothesis.

A **one-tailed test** is used when the alternative hypothesis specifies a direction of the effect. This means that the test checks whether a parameter is either **greater than or less than** a certain value.

Example:

A company wants to test whether the average salary of employees is **greater than ₹50,000**.

Hypotheses:

$$H_0: \mu \leq 50,000$$

$$H_1: \mu > 50,000$$

Since the test checks only one direction (greater than), it is a **one-tailed test**.

A **two-tailed test** is used when the alternative hypothesis does not specify a direction. It tests whether the parameter is **different from** a certain value, either greater or smaller.

Example:

A researcher wants to check whether the average salary of employees is **different from ₹50,000**.

Hypotheses:

$$H_0: \mu = 50,000$$

$$H_1: \mu \neq 50,000$$

Since the test checks both directions (greater or less), it is a **two-tailed test**.

Thus, the main difference is that a one-tailed test examines **one direction**, while a two-tailed test examines **both directions**.

Question 5: A company claims that the average time to resolve a customer complaint is 10 minutes. A random sample of 9 complaints gives an average time of 12 minutes and a standard deviation of 3 minutes. At $\alpha = 0.05$, test the claim.

A company claims that the average time to resolve a customer complaint is **10 minutes**. A random sample of **9 complaints** shows an average time of **12 minutes** with a standard deviation of **3 minutes**. The significance level is **0.05**.

Step 1: State the hypotheses

$$H_0: \mu = 10$$

$$H_1: \mu \neq 10$$

Step 2: Choose the test

Since the sample size is **less than 30** and population standard deviation is not known, we use the **t-test**.

The test statistic is:

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

Step 3: Calculate the test statistic

Sample mean = 12

Population mean = 10

Standard deviation = 3

Sample size = 9

$$t = (12 - 10) / (3 / \sqrt{9})$$

$$t = 2 / (3 / 3)$$

$$t = 2$$

Step 4: Determine the critical value

Degrees of freedom:

$$df = n - 1 = 9 - 1 = 8$$

For $\alpha = 0.05$ (**two-tailed**), the critical t-value ≈ 2.306

Step 5: Decision

Since **calculated $t = 2$** is **less than critical value 2.306**, we **fail to reject the null hypothesis**.

Step 6: Conclusion

There is not enough statistical evidence to conclude that the average complaint resolution time is different from **10 minutes**. Therefore, the company's claim cannot be rejected.

Question 6: When should you use a Z-test instead of a t-test?

A **Z-test** is used in hypothesis testing when certain conditions are satisfied.

A Z-test should be used when:

1. The **sample size is large ($n \geq 30$)**.
2. The **population standard deviation (σ) is known**.
3. The data follows a **normal distribution** or the sample size is sufficiently large for the Central Limit Theorem to apply.

On the other hand, a **t-test** is used when:

1. The **sample size is small ($n < 30$)**.
2. The **population standard deviation is unknown**.
3. The standard deviation must be estimated using the **sample standard deviation**.

The t-distribution is similar to the normal distribution but has **heavier tails**, which accounts for the additional uncertainty caused by using the sample standard deviation instead of the population standard deviation.

As the sample size increases, the t-distribution gradually approaches the normal distribution.

Therefore, Z-tests are generally used for **large samples**, while t-tests are more appropriate for **small sample sizes**.

Question 7: The productivity of 6 employees was measured before and after a training program.

Employee	Before	After
1	50	55
2	60	65
3	58	59
4	55	58
5	62	63
6	56	59

At $\alpha = 0.05$, test if the training improved productivity.

The productivity of **6 employees** was measured **before and after training**. Since the same employees are measured twice, we use a **Paired t-test**.

Step 1: State the Hypotheses

Null Hypothesis (H_0):

Training did not improve productivity.

$$H_0: \mu d = 0$$

Alternative Hypothesis (H_1):

Training improved productivity.

$$H_1: \mu d > 0$$

where $d = \text{After} - \text{Before}$

Step 2: Calculate the Differences

Employee	Before	After	Difference (d)
1	50	55	5
2	60	65	5
3	58	59	1
4	55	58	3
5	62	63	1
6	56	59	3

$$\sum d = 18$$

Mean difference:

$$\bar{d} = \frac{18}{6} = 3$$

Step 3: Calculate Standard Deviation

d	$d - \bar{d}$	$(d - \bar{d})^2$
5	2	4
5	2	4
1	-2	4
3	0	0
1	-2	4
3	0	0

$$\sum (d - \bar{d})^2 = 16$$

Standard deviation:

$$s_d = \sqrt{\frac{16}{5}} = 1.789$$

Step 4: Test Statistic

The paired t-test formula is:

$$t = \frac{\bar{d}}{sd/\sqrt{n}}$$

Substitute values:

$$t = \frac{3}{1.789\sqrt{n}}$$

$$t = \frac{3}{0.73}$$

$$t \approx 4.10$$

Step 5: Critical Value

Significance level:

$$\alpha = 0.05$$

Degrees of freedom:

$$df = n - 1 = 6 - 1 = 5$$

Critical value (one-tailed):

$$t_{0.05,5} = 2.015$$

Step 6: Decision

Since

$$t_{\text{calculated}} = 4.10 > 2.015$$

We reject the null hypothesis.

Step 7: Conclusion

There is sufficient statistical evidence at the **5% significance level** to conclude that the **training program significantly improved employee productivity**.

Question 8: A company wants to test if product preference is independent of gender

Gender	Product A	Product B	Total
Male	30	20	50
Female	10	40	50
Total	40	60	100

At $\alpha = 0.05$, test independence

A company wants to test whether **product preference depends on gender**. This is tested using the **Chi-Square Test of Independence**.

Step 1: State the Hypotheses

Null Hypothesis (H_0):

Product preference is **independent of gender**.

Alternative Hypothesis (H_1):

Product preference is **dependent on gender**.

Step 2: Observed Frequencies

Gender	Product A	Product B	Total
Male	30	20	50
Female	10	40	50
Total	40	60	100

Step 3: Expected Frequencies

Expected frequency formula:

$$E = \frac{(\text{Row Total})(\text{Column Total})}{\text{Grand Total}}$$

Expected values:

Male-A:

$$E = \frac{50 \times 40}{100} = 20$$

Male-B:

$$E = \frac{50 \times 60}{100} = 30$$

Female-A:

$$E = 20$$

Female-B:

$$E = 30$$

Expected table:

Gender	Product A	Product B
Male	20	30
Female	20	30

Step 4: Chi-Square Statistic

The formula is:

$$X^2 = \sum \frac{(O-E)^2}{E}$$

Calculate each value:

Male-A:

$$\frac{(30-20)^2}{20} = 5$$

Male-B:

$$\frac{(20-30)^2}{30} = 3.33$$

Female-A:

$$\frac{(10-20)^2}{20} = 5$$

Female-B:

$$\frac{(40-30)^2}{30} = 3.33$$

Total:

$$X^2 = 5 + 3.33 + 5 + 3.33 = 16.66$$

Step 5: Critical Value

Degrees of freedom:

$$df = (r-1)(c-1)$$

$$df = (2-1)(2-1) = 1$$

At $\alpha = 0.05$, the critical value is:

$$X^2_{0.05,1} = 3.84$$

Step 6: Decision

Since

$$16.66 > 3.84$$

We reject the null hypothesis.

Step 7: Conclusion

There is sufficient statistical evidence to conclude that **product preference is not independent of gender**. This means **gender influences product choice**.